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Task-2 Hand-Tracing Task

Initial board configuration:

- d1 = 0 : 0 mod 4 = 0, so r1 = 0 + 1 = 1
- d2 = 1 : 1 mod 4 = 1, so r2 = 1 + 1 = 2
- d3 = 8 : 8 mod 4 = 0, so r3 = 0 + 1 = 1
- d4 = 5 : 5 mod 4 = 1, so r4 = 1 + 1 = 2

starting state: [r1, r2, r3, r4] = **[1, 2, 1, 2]**

➢ Simple Hill Climbing:

Step 1: **[1,2,1,2] h = 5**

#	Neighbor state	h	Better?
1	[2,2,1,2]	5	No
2	[3,2,1,2]	4	Yes (first improvement)
3	[4,2,1,2]	3	Yes (last improvement)

Step 2: **[4,2,1,2] h = 3**

#	Neighbor state	h	Better?
1	[1,2,1,2]	5	No
2	[2,2,1,2]	5	No
3	[3,2,1,2]	5	No

4	[4,1,1,2]	2	Yes (first improvement)
5	[4,3,1,2]	2	Yes (better but not first)
6	[4,4,1,2]	3	no

Simple HC final outcome

Item	**Value**
Initial board	[1, 2, 1, 2], h = 5
After Step 1	[4, 2, 1, 2], h = 3
After Step 2	[4, 1, 1, 2], h = 2
Valid 4-Queens solution?	No (still 2 attacking pairs)
Outcome after 2 steps	Not a solution, but not yet stuck better neighbors still exist if we continued

➢ Simulated Annealing one step:

State = [1,2,1,2], h = 5, T = 10

#	Neighbor state	h	Better?
1	[2,2,1,2]	5	sideways move, not better and not worse

Quantity	Value
Current board	[1, 2, 1, 2]
Candidate board	[2, 2, 1, 2]
h_current	5
h_new	5
Delta = h_current - h_new	0
Temperature (T)	10.0
Acceptance probability (p = e^(Delta/T)	e^(0/10) = 1

Simulated Annealing (final outcome):

Item	Value
Start	[1, 2, 1, 2], h=5, T=10
After 1 SA step	[2, 2, 1, 2], h=5, T=9.9
Move type	Sideways (not better or worse)
Any worse move accepted?	No, in this first step there was no neighbor with (h > 5) only sideways moves were possible
Valid solution after 1 step?	No (still 5 attacking pairs)
Outcome	Algorithms remain in a conflicted state and need more steps and early high temperature

Task-3. Comparative Analysis

3.1 - 20 initial 8-Queens boards (ID = 140185)

1: [3, 3, 2, 4, 3, 3, 5, 7]

2: [5, 4, 7, 2, 2, 0, 3, 6]

3: [6, 6, 4, 7, 2, 1, 4, 3]

4: [0, 7, 7, 1, 1, 7, 3, 6]

5: [6, 0, 0, 0, 7, 3, 5, 3]

6: [2, 4, 3, 0, 7, 2, 2, 0]

7: [3, 4, 6, 5, 6, 1, 4, 0]

8: [0, 7, 5, 4, 1, 1, 4, 2]

9: [7, 2, 0, 4, 1, 7, 0, 6]

10: [0, 6, 2, 2, 5, 7, 5, 0]

11: [4, 5, 2, 2, 5, 4, 1, 6]

12: [2, 2, 0, 1, 6, 5, 6, 5]

13: [2, 4, 2, 2, 3, 3, 3, 5]

14: [6, 6, 5, 5, 2, 4, 6, 7]

15: [1, 6, 5, 1, 0, 6, 2, 0]

16: [7, 1, 1, 5, 7, 5, 0, 0]

17: [7, 2, 4, 3, 5, 0, 4, 2]

18: [1, 3, 7, 5, 0, 3, 0, 4]

19: [6, 3, 0, 4, 1, 5, 7, 7]

20: [5, 6, 7, 3, 3, 5, 7, 3]

3.2 - Detailed per-run results (20 runs per algorithm)

Note: If success then 1 , in not success then 0

- **Algorithm: Simple HC**

Board #	Success	Steps	Time (ms)	Final h	Failure Mode
1	0	8	3.863	1	Stuck in local minimum
2	0	7	2.998	1	Stuck in local minimum
3	0	6	2.246	2	Stuck in local minimum
4	0	4	1.558	1	Stuck in local minimum
5	0	2	1.064	3	Stuck in local minimum
6	0	5	1.958	1	Stuck in local minimum
7	0	4	1.583	1	Stuck in local minimum
8	0	0	0.310	3	Stuck in local minimum
9	0	3	2.023	1	Stuck in local minimum
10	0	3	1.281	3	Stuck in local minimum
11	0	7	2.523	1	Stuck in local minimum
12	0	9	3.685	2	Stuck in local minimum
13	0	6	2.207	2	Stuck in local minimum
14	0	4	1.561	1	Stuck in local minimum
15	0	7	2.558	2	Stuck in local minimum
16	0	3	1.252	2	Stuck in local minimum
17	0	4	1.735	2	Stuck in local minimum
18	0	3	1.803	1	Stuck in local minimum
19	0	1	1.153	1	Stuck in local minimum
20	0	8	2.935	1	Stuck in local minimum

- **Algorithm: Steepest HC**

Board #	Success	Steps	Time (ms)	Final h	Failure Mode
1	0	3	1.298	2	Stuck in local minimum
2	0	2	0.964	2	Stuck in local minimum
3	0	4	1.873	2	Stuck in local minimum
4	0	3	1.263	1	Stuck in local minimum
5	0	2	0.969	2	Stuck in local minimum
6	1	5	1.582	0	-
7	0	2	0.996	2	Stuck in local minimum
8	0	0	0.333	3	Stuck in local minimum
9	1	3	1.105	0	-
10	0	3	1.277	1	Stuck in local minimum
11	0	4	1.600	1	Stuck in local minimum
12	0	3	1.285	3	Stuck in local minimum
13	0	2	1.043	2	Stuck in local minimum
14	0	3	1.636	1	Stuck in local minimum
15	0	5	1.924	1	Stuck in local minimum
16	0	3	1.253	2	Stuck in local minimum
17	0	2	1.238	2	Stuck in local minimum
18	0	3	1.292	1	Stuck in local minimum
19	0	1	0.642	1	Stuck in local minimum
20	0	4	1.609	1	Stuck in local minimum

- **Algorithm: Stochastic HC**

Board #	Success	Steps	Time (ms)	Final h	Failure Mode
1	0	5	1.989	1	Stuck in local minimum
2	0	4	1.745	1	Stuck in local minimum
3	0	7	2.606	2	Stuck in local minimum
4	0	3	1.303	3	Stuck in local minimum
5	0	4	2.261	2	Stuck in local minimum
6	1	7	2.668	0	-
7	0	3	1.324	3	Stuck in local minimum
8	0	0	0.330	3	Stuck in local minimum
9	0	3	1.316	1	Stuck in local minimum
10	0	3	1.491	2	Stuck in local minimum
11	0	6	2.227	1	Stuck in local minimum
12	0	7	2.915	1	Stuck in local minimum
13	0	4	2.744	2	Stuck in local minimum
14	0	4	1.790	2	Stuck in local minimum
15	0	6	2.566	1	Stuck in local minimum
16	0	4	1.619	2	Stuck in local minimum
17	0	5	1.936	1	Stuck in local minimum
18	0	3	1.637	2	Stuck in local minimum
19	0	1	0.729	1	Stuck in local minimum
20	1	7	3.193	0	-

- **Algorithm: Random-Restart HC**

Board #	Success	Steps	Time (ms)	Final h	Failure Mode
1	1	40	17.133	0	-
2	1	22	9.498	0	-
3	1	21	8.985	0	-
4	1	16	6.520	0	-
5	1	6	3.474	0	-
6	1	5	1.584	0	-
7	1	14	5.466	0	-
8	1	23	9.487	0	-
9	1	3	1.235	0	-
10	1	21	9.269	0	-
11	1	16	6.120	0	-
12	1	29	13.383	0	-
13	1	8	3.538	0	-
14	1	13	6.578	0	-
15	1	64	26.846	0	-
16	1	31	12.895	0	-
17	1	22	9.376	0	-
18	1	12	4.509	0	-
19	1	10	3.878	0	-
20	1	16	6.110	0	-

- **Algorithm: Simulated Annealing**

Board #	Success	Steps	Time (ms)	Final h	Failure Mode
1	0	247	13.254	1	Temperature too low to escape poor regions
2	1	219	7.014	0	-
3	0	227	13.638	1	Temperature too low to escape poor regions
4	1	64	1.833	0	-
5	1	239	13.032	0	-
6	1	200	5.565	0	-
7	1	237	8.134	0	-
8	1	229	9.656	0	-
9	0	247	14.129	1	Temperature too low to escape poor regions
10	1	244	12.084	0	-
11	0	226	14.284	1	Temperature too low to escape poor regions
12	0	256	14.283	1	Temperature too low to escape poor regions
13	1	194	6.050	0	-
14	1	201	7.179	0	-
15	1	235	9.948	0	-
16	1	189	5.316	0	-
17	1	229	10.812	0	-
18	1	224	7.263	0	-
19	1	227	10.707	0	-

20	0	235	13.471	1	Temperature too low to escape poor regions

3.3 - Performance Summary Table

Algorithm	**Success Rate (%)**	**Average Steps**	**Time Taken (ms)**	**Typical Failure Mode**
Simple HC	0	4.70	1.840	Stuck in local minimum
Steepest HC	10	2.85	1.278	Stuck in local minimum
Stochastic HC	10	4.30	1.726	Stuck in local minimum
Random-Restart HC	100	19.60	7.903	not applicable all runs succeeded
Simulated Annealing	70	218.45	9.894	Temperatures are too low to escape poor regions